LAB 5: STATIC CODE ANALYSIS

Telecom Tower Maintenance Tracker

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1. Summary of Results

Tool	Before	After
Pylint	5.14/10	10.00/10
Bandit	B105 hardcoded password; B608 possible SQL injection; B301 pickle use	No issues identified
Flake8	E501 long lines; E303 excess blank lines; E225 operator whitespace	No violations reported (empty report)

The final program output was ['Ridge-A'] high 2 1. The final Pylint score was 10.00/10, Bandit reported “No issues identified,” and Flake8 produced an empty report, as shown in the terminal screenshots.

2. Issue Documentation Table

Line numbers refer to the original file as shown in the initial tool reports; line numbers changed when the program was rewritten.

Issue / tool code	Type	Original line(s)	Description	Fix approach / outcome
Hardcoded password (B105)	Security	6	Password-like value 'tower-noc-77' was stored in source code and could expose a credential if shared.	Removed the unused NOC_PASSWORD constant; it was not used by the program.
Possible SQL injection (B608)	Security	11	The INSERT query was built with string formatting rather than binding the input values.	Changed to SQLite placeholders (?, ?, ?) and passed the values as a tuple.
Unsafe pickle use (B301)	Security	43	snapshot() used pickle.loads(pickle.dumps(...)) unnecessarily; pickle deserialization can be unsafe with untrusted data.	Removed pickle and returned fetched SQLite rows directly.
Long lines (E501 / C0301)	Style / maintainability	5, 11, 18, 35, 43	Several source lines exceeded the 79-character limit.	Split SQL statements and expressions across lines; final Flake8 report was empty.
Excess blank lines (E303)	Style	25	Too many consecutive blank lines appeared between functions.	Removed excess blank lines.
Missing operator whitespace (E225)	Style	50	total=len(snapshot()) lacked spaces around the assignment operator.	Changed to total = len(snapshot()).
Missing docstrings (C0114 / C0116)	Code quality	1, 10, 15, 25, 34, 42, 46	The module and functions did not consistently explain their purpose.	Added module and function docstrings.
Non-snake_case names (C0103)	Maintainability	10, 15, 34	Names such as addTower, serviceTower, and riskLevel did not follow Python naming conventions.	Renamed functions to add_tower, service_tower, and risk_level, updating call sites.
Broad exception handling (W0718)	Reliability	20	Catching Exception could hide unrelated errors and complicate debugging.	Changed the handler to catch sqlite3.Error.
Global statement (W0603)	Maintainability	16	The maintenance function modified a module-level counter with global.	service_tower() now returns success/failure; main() tracks service_count locally.
Missing-row handling	Robustness improvement	riskLevel function	The original function accessed row[0] without checking whether a row existed.	Added a None check and return 'unknown' for a nonexistent tower ID.

4. Verification Evidence

- Original Pylint score: 5.14/10.
- Final Pylint score: 10.00/10.
- Final Bandit result: No issues identified.
- Final Flake8 report: empty; no style violations were printed.
- Final program output: ['Ridge-A'] high 2 1.
- Before-and-after report files: pylint_before.txt, pylint_after.txt, bandit_before.txt, bandit_after.txt, flake8_before.txt, flake8_after.txt.
- Git commits visible on the fork: eb2705b, 332f61f, and 4beab38.